VECTOR VALUED FUNCTIONS

Vector valued functions of a real variable

If $\bar{r}(t)$ is a vector valued function then $\bar{r}(t)$ can be represented in terms of component and each component is a fn of real values of 't'.

$$\bar{r}(t) = \langle x(t), y(t), z(t) \rangle$$

$$\bar{r}(t) = x(t)\,\hat{i} + y(t)\,\hat{j} + z(t)\,\hat{k}$$

Q. Express the parametric equations $x = \frac{1}{t}$, $y = \sqrt{t}$, and $z = \sin^{-1} t$ as a single vector equation

$$\bar{r}(t) = \frac{1}{t}\,\hat{i} + \sqrt{t}\,\hat{j} + \sin^{-1} t\,\hat{k}$$

Calculus of vector valued functions.

I Limits & Continuity

For a vector valued function

$\bar{r}(t) = \langle f(t), g(t), h(t) \rangle$

$\lim_{t \to a} \bar{r}(t) = \left\langle \lim_{t\to a} f(t), \lim_{t \to a} g(t), \lim_{t\to a} h(t) \right\rangle$

provided the limits of all components exist. If any of the limits indicated on the right hand side fail to exist then $\lim_{t\to a} \bar{r}(t)$ doesnot exist

Continuity

$\bar{r}(t)$ is continuous when f, g, h are continuous

1) Find the limits of the vector valued fn.

$\bar{r}(t) = t^2\hat{i} + e^t\hat{j} - 2\cos\pi t\,\hat{k}$, find $\lim_{t\to 0} \bar{r}(t)$

Ans:- $\lim_{t\to 0} \bar{r}(t) = \lim_{t\to 0} t^2\hat{i} + \lim_{t\to 0} e^t\hat{j} - \lim_{t\to 0} 2\cos\pi t\,\hat{k}$

$= \underline{0\hat{i} + \hat{j} - 2\hat{k}} = \hat{j} - 2\hat{k}$

2) Find $\lim_{t\to\infty} \left\langle \frac{t^3+1}{4t^3+2}, \frac{1}{t} \right\rangle$

$$\left\langle \lim_{t\to\infty} \frac{t^3+1}{4t^3+2}, \lim_{t\to\infty} \frac{1}{t} \right\rangle$$

$$= \left\langle \lim_{t\to\infty} \frac{t^3(1+\frac{1}{t^3})}{t^3(4+\frac{2}{t^3})}, \frac{1}{\infty} \right\rangle$$

$$= \left\langle \frac{1}{4}, 0 \right\rangle = \underline{\underline{\frac{1}{4}\hat{i}}}$$

3) $\lim_{t\to 0} \left\langle 2t, \frac{\sin t}{t} \right\rangle$

$$= \left\langle \lim_{t\to 0} 2t, \lim_{t\to 0} \frac{\sin t}{t} \right\rangle$$

$$= \left\langle 0, \lim_{t\to 0} \frac{\cos t}{1} \; \text{(L-H rule)} \right\rangle$$

$$= \langle 0, 1 \rangle = \underline{\hat{j}}$$

4) Find $\lim_{t\to 0} \left\langle e^{2t}+5, t^2-2t-3, \frac{1}{t} \right\rangle$

$= \langle 6, -3,$ limit of $\frac{1}{t}$ doesnot exist$\rangle$

$\therefore$ Limit of the above vector valued fn doesnot exist.

H.W 5) $\lim_{t\to 3} \langle t^2+2t \rangle = \langle 9, 6 \rangle$

H.W 6) $\lim_{t\to \pi/4} \langle \cos t, \sin t \rangle = \left\langle \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right\rangle$

Problems on continuity

Note:- Trigonometric fns like, sine, cosine exponential function, modulus fn, polynomial functions etc are continuous everywhere.

Q Determine whether $\bar{r}(t)$ is continuous at $t=0$, explain your reasoning.

1) $\bar{r}(t) = 3\sin t\,\hat{i} + 3t\,\hat{j}$

Ans Each of the components are continuous everywhere $\therefore$ continuous at $t=0$

ie $\bar{r}(t)$ is continuous at $t=0$

2. $\bar{r}(t) = t^2\hat{i} + \frac{\cos t}{t}\hat{j} + t\hat{k}$

Ans:- Second component $\frac{\cos t}{t}$ is not continuous at $t=0$. $\therefore \bar{r}(t)$ is not continuous at $t=0$

3. Determine for what values of 't' the vector valued function $\bar{r}(t) = \langle e^{5t}, \ln(t+1), \cos t \rangle$ is continuous.

Ans:- $\ln(t+1)$ is continuous at $t+1>0$

$t>-1$

$\therefore \bar{r}(t)$ is continuous at $t>-1$

II Derivatives of vector valued functions.

Let $\bar{r}(t) = \langle f(t), g(t), h(t) \rangle$. The derivati
$\bar{r}'(t)$ is defined only when all the compon
are differentiable for some values of 't'.

i) If $\bar{r}(t) = t^2\hat{i} + e^t\hat{j} - 2\cos \pi t \hat{k}$ find $\bar{r}'(t)$

Ans:-

$$\bar{r}'(t) = \frac{d}{dt} t^2 \hat{i} + \frac{d}{dt} e^t \hat{j} - \frac{d}{dt} 2\cos \pi t \hat{k}$$

$$= \underline{2t\hat{i} + e^t\hat{j} + 2\pi \sin \pi t \hat{k}}$$

The norm (Magnitude) of a vector v ($\|\bar{v}\|$)

If $\bar{v} = v_1\hat{i} + v_2\hat{j} + v_3\hat{k}$

then $\|\bar{v}\| = \sqrt{v_1^2 + v_2^2 + v_3^2}$

Integration of a vector valued function.

The integral of a vector valued function exist only when integral of all the components exist.

1) Evaluate $\int (5\hat{i} + 4t\hat{j})\,dt$

Ans- $5t\hat{i} + \frac{4t^2}{2}\hat{j} + c_1\hat{i} + c_2\hat{j} = (5t + c_1)\hat{i} + (2t^2 + c_2)\hat{j}$

2) Calculate $\int \langle \cos t, \sin t \rangle\,dt$

$\langle \sin t, -\cos t \rangle + \langle c_1, c_2 \rangle$

$= \langle \sin t + c_1, -\cos t + c_2 \rangle$

3) Evaluate $\int_0^{\pi/2} \langle \cos 2t, \sin 2t \rangle\,dt$

$= \left\langle \int_0^{\pi/2} \cos 2t\,dt, \int_0^{\pi/2} \sin 2t\,dt \right\rangle$

$= \left\langle \left(\frac{\sin 2t}{2}\right)_0^{\pi/2}, \left(\frac{-\cos 2t}{2}\right)_0^{\pi/2} \right\rangle = \left\langle \frac{1}{2}\sin\pi - \sin 0, \frac{-1}{2}(\cos\pi + \cos 0) \right\rangle$

$$\left\langle 0, \frac{1}{2}(+1+1) \right\rangle = \langle 0, 1 \rangle = \hat{j}$$

4. Evaluate $\int_1^9 \left(t^{1/2}\hat{i} + t^{-1/2}\hat{j}\right) dt$

$$= \left(\frac{t^{3/2}}{3/2}\hat{i} + \frac{t^{1/2}}{1/2}\hat{j}\right)_1^9$$

$$= \frac{2}{3}\left((9^{3/2} - 1)\right)\hat{i} + 2\left(9^{1/2} - 1\right)\hat{j}$$

$$= \frac{2}{3}(27-1)\hat{i} + 4\hat{j}$$

$$= \frac{52}{3}\hat{i} + 4\hat{j}$$

Dot product (Scalar product)

$\bar{a}$ & $\bar{b}$ are two vectors in 3D space such that

$\bar{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$, $\bar{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$

then $\bar{a}\cdot\bar{b} = a_1b_1 + a_2b_2 + a_3b_3$

It is a scalar quantity.

Note
$\hat{i}\cdot\hat{i} = 1$
$\hat{j}\cdot\hat{j} = 1$
$\hat{k}\cdot\hat{k} = 1$
$\hat{i}\cdot\hat{j} = 0$
$\hat{j}\cdot\hat{k} = 0$
$\hat{k}\cdot\hat{i} = 0$

Cross Product (Vector product)

$$\bar{a}\times\bar{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$ Expand this determinant.

It is a vector quantity.

Eg:-1) Find $\bar{r}_1\cdot\bar{r}_2$ and $\bar{r}_1\times\bar{r}_2$ where $r_1(t) = 2t\hat{i} + 3t^2\hat{j} + t^3\hat{k}$ $r_2(t) = t^4\hat{k}$

Ans:- $\bar{r}_1\cdot\bar{r}_2 = (2t\hat{i} + 3t^2\hat{j} + t^3\hat{k})\cdot(0\hat{i} + 0\hat{j} + t^4\hat{k}) = 0 + 0 + t^3t^4 = \underline{\underline{t^7}}$

$$\bar{r}_1\times\bar{r}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2t & 3t^2 & t^3 \\ 0 & 0 & t^4 \end{vmatrix} = \hat{i}(3t^6 - 0) - \hat{j}(2t^5 - 0) + \hat{k}(0)$$

$$= \underline{\underline{3t^6\hat{i} - 2t^5\hat{j}}}$$

2) Find $\bar{a}\times\bar{b}$ where $\bar{a} = 2\hat{i} + 3\hat{j}$, $\bar{b} = 4\hat{i} + 7\hat{j}$

$$\bar{a}\times\bar{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 0 \\ 4 & 7 & 0 \end{vmatrix} = \hat{i}(0) - \hat{j}(0) + \hat{k}(14 - 12) = \underline{\underline{2\hat{k}}}$$

Motion along a Curve

If $\bar{r}(t)$ is the position function of a moving particle.

Then instantaneous velocity at that instant

$$\boxed{\bar{v}(t) = \bar{r}'(t) = \frac{d}{dt}\bar{r}(t)}$$

Instantaneous Acceleration $\bar{a}(t) = \bar{r}''(t) = \frac{d^2 r}{dt^2}$

Instantaneous speed $= \|\bar{v}(t)\|$

1. A particle moves along a circular path in such a way that its x and y co-ordinates at time 't' are
$x = 2\cos t$, $y = 2\sin t$

a) Find the instantaneous velocity & speed of the particle at time t.

b) Show that at each instant, the acceleration vector is $\perp r$ to the velocity vector

Ans- $\bar{r}(t) = 2\cos t\, \hat{i} + 2\sin t\, \hat{j}$

a) Velocity $\bar{v}(t) = -2\sin t\, \hat{i} + 2\cos t\, \hat{j}$

Speed $\|\bar{v}(t)\| = \sqrt{4\sin^2 t + 4\cos^2 t} = \underline{2}$

b) Acceleration $\bar{a}(t) = -2\cos t\, \hat{i} - 2\sin t\, \hat{j}$

To show that velocity vector is $\perp r$ to acceleration vector prove that $\bar{v}\cdot\bar{a}=0$

$\bar{v}\cdot\bar{a} = 4\cos t\sin t - 4\cos t\sin t = 0$

$\therefore$ $\bar{v}$ is $\perp$r to $\bar{a}$

2) Find the velocity acceleration & speed of a particle moving along the curve

$x = 1+3t,\ y = 3-4t,\ z = 7+3t$ at $t=2$

Ans:- $\bar{r}(t) = \langle 1+3t,\ 3-4t,\ 7+3t\rangle$

$\bar{v}(t)\ \bar{r}'(t) = \langle 3,\ -4,\ 3\rangle$

$\therefore \bar{v}(2) = \langle 3,-4,3\rangle$

$\bar{a}(t) = \bar{r}''(t) = 0$

Speed $= \|v\| = \sqrt{9+16+9} = \sqrt{34}$

H.W 3) Find the velocity, speed & acceleration at the given time t of a particle moving along the given curve.

$\bar{r}(t) = e^t\sin t\,\hat{i} + e^t\cos t\,\hat{j} + t k$ at $t=\pi$

Ans $\bar{v}(t) = \langle e^t\cos t + e^t\sin t,\ -e^t\sin t + e^t\cos t,\ 1\rangle$

$\bar{v}(\pi) = \langle -e^{\pi},\ e^{\pi},\ 1\rangle$

$\|\bar{v}\| = \sqrt{e^{2\pi}+e^{2\pi}+1} = \sqrt{1+2e^{2\pi}}$

$\bar{v}(t) = \langle e^t(\cos t+\sin t),\ e^t(\cos t-\sin t),\ 1\rangle$

$$\bar{a}(t) = \langle e^{t}(-\sin t + \cos t) + e^{t}(\cos t + \sin t),$$
$$e^{t}(-\sin t - \cos t) + e^{t}(\cos t - \sin t), 0 \rangle$$

$$\bar{a}(t) = \langle 2e^{t}\cos t, -2e^{t}\sin t, 0 \rangle$$

$$a(\pi) = \langle 2e^{\pi}\cos\pi, -2e^{\pi}\sin\pi, 0 \rangle$$

$$= \langle -2e^{\pi}, 0, 0 \rangle$$

$$= \underline{-2e^{\pi}\hat{i}}$$

4) Use the given information to find the position & velocity vectors of the particle.
$\bar{a}(t) = -\cos t\,\hat{i} - \sin t\,\hat{j}$ $V(0) = i$, $\bar{r}(0) = 2\hat{j}$

Ans: $\bar{V}(t) = \int \bar{a}(t)\, dt$

$$= \int (-\cos t\, i - \sin t\, j)\, dt$$

$$= (-\sin t)i + (\cos t\, j) + C_1 i + C_2 j$$

$$= (-\sin t + C_1)i + (\cos t + C_2)\hat{j}$$

given that $\bar{V}(0) = i$

ie $\bar{V}(0) = (-\sin t + C_1)i + (\cos t + C_2)\hat{j}$

$i = (-\sin\theta + C_1)\hat{i} + (\cos\theta + C_2)\hat{j}$

$i = \cancel{-\sin t}\ C_1 i + (i + C_2)\hat{j}$

$C_1 = 1, \quad C_2 = -1$

$\therefore \bar{v}(t) = \underline{(-\sin t + 1)\hat{i} + (\cos t - 1)\hat{j}}$

$\bar{r}(t) = \int (-\sin t + 1)\hat{i} + (\cos t - 1)\hat{j}\ dt$

$= (\cos t + t)\hat{i} + (\sin t - t)\hat{j} + C_3 i + C_4 \hat{j}$

$= \underline{(\cos t + t + C_3)\hat{i} + (\sin t - t + C_4)\hat{j}}$

given that $\bar{r}(0) = 2\hat{j}$

$\bar{r}(0) = (1 + C_3)\hat{i} + C_4\hat{j}$

$2\hat{j} = (1 + C_3)\hat{i} + C_4\hat{j} \quad \therefore C_4 = 2, \ C_3 = -1$

$\therefore \bar{r}(t) = \underline{(\cos t + t - 1)\hat{i} + (\sin t - t + 2)\hat{j}}$

H.W

5) Use the given information to find the position & velocity vectors of the particle

$\bar{a}(t) = \hat{i} + e^{-t}\hat{j} \quad \bar{v}(0) = 2\hat{i} + \hat{j} \quad \bar{r}(0) = \hat{j} - \hat{i}$

Ans:- $\bar{a}(t) = \hat{i} + e^{-t}\hat{j}$

$\bar{v}(t) = \int (i + e^{-t} j)\,dt$

$= \cancel{\phi}\ ti + \frac{e^{-t}}{-1} j + C_1 i + C_2 \hat{j}$

$= (t + C_1)\hat{i} + (-e^{-t} + C_2)\hat{j}$

$\bar{v}(0) = (0 + C_1)\hat{i} + (-1 + C_2)\hat{j}$

$2\hat{i} + \hat{j} = C_1\hat{i} + (-1 + C_2)\hat{j} \quad \therefore C_1 = 2, \ C_2 = 2$

$\bar{v}(t) = (t+2)\hat{i} + (-e^{-t}+2)\hat{j}$

$\bar{r}(t) = \int (t+2)\hat{i} + (-\bar{e}^{t}+2)\hat{j} \; dt$

$= \left(\frac{t^2}{2} + 2t + c_3\right)\hat{i} + +(\bar{e}^{t} + 2t + c_4)\hat{j}$

$\bar{r}(0) = c_3\hat{i} + (1 + c_4)\hat{j}$

$\hat{j} - \hat{i} = c_3\hat{i} + (1+c_4)\hat{j}$

$c_4 = 0 \quad , \quad c_3 = -1$

$\therefore \bar{r}(t) = \left(\frac{t^2}{2} + 2t - 1\right)\hat{i} + (\bar{e}^{t} + 2t)\hat{j}$

6) $\bar{a}(t) = \sin t\,\hat{i} + \cos t\,\hat{j} + e^{t}\hat{k} \quad v(0) = \hat{k}, \; r(0) = \hat{j}$

Ans $\bar{v}(t) = (1-\cos t)\hat{i} + \sin t\,\hat{j} + e^{t}\hat{k}$

$\bar{r}(t) = (t - \sin t)\hat{i} + (2 - \cos t)\hat{j} + (e^{t} - 1)\hat{k}$

Distance and displacement travelled

Distance → Scalar; displacement → Vector

If a particle travels along a curve C in 2 space or 3 space,
the displacement of the particle during the time $t_1 < t < t_2$ is denoted by $\Delta \bar{r}$.

ie Displacement $\boxed{\Delta \bar{r} = \bar{r}(t_2) - \bar{r}(t_1)}$

OR $\boxed{\Delta \bar{r} = \int_{t_1}^{t_2} \bar{V}(t)\, dt}$

Distance Travelled $\boxed{S = \int_{t_1}^{t_2} \| \bar{V}(t) \|\, dt}$

Eg:- 1) Suppose that a particle moves along a circular helix in 3-space so that the position vector at time is $\bar{r}(t) = (4\cos\pi t)\hat{i} + (4\sin\pi t)\hat{j} + t\hat{k}$. Find the distance & displacement of the particle in $1 < t < 5$

Ans: $V(t) = \bar{r}'(t)$

$= (-4\sin\pi t \cdot \pi)\hat{i} + (4\cos\pi t \cdot \pi)\hat{j} + \hat{k}$

$\|V(t)\| = \sqrt{16\pi^2 \sin^2\pi t + 16\pi^2 \cos\pi t + 1} = \sqrt{16\pi^2 + 1}$

Disance $S = \int_1^5 \sqrt{16\pi^2+1}\, dt = \sqrt{16\pi^2+1}\,(t)_1^5$

$= \sqrt{16\pi^2+1}\,(5-1)$

$= 4\sqrt{16\pi^2+1}$

Displacement $= \Delta\bar{r} = \bar{r}(t_2) - \bar{r}(t_1)$

$= \bar{r}(5) - \bar{r}(1)$

$= (4\cos 5\pi\, \hat{i} + 4\sin 5\pi\, \hat{j} + 5\hat{k})$

$-(4\cos\pi\, \hat{i} + 4\sin\pi\, \hat{j} + \hat{k})$

$= (-4\hat{i} + 5\hat{k}) - (-4\hat{i} + \hat{k})$

$= 4\hat{k}$

H.W Find the displacement and the distance travelled over the indicated time interval

$\bar{r}(t) = (1 - 3\sin t)\hat{i} + 3\cos t\, \hat{j}; \quad 0 < t < \pi$

Ans :- Displacement $\Delta\bar{r} = -6\hat{j}$ (Check your answers)

Distance $S = 3\pi$

The Gradient (Vector differential operator) (∇ del)

$$\nabla \equiv \frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}$$

Gradient of a function $f(x,y,z)$

$$\nabla f = \left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}\right)f$$

$$= \frac{\partial f}{\partial x}\hat{i} + \frac{\partial f}{\partial y}\hat{j} + \frac{\partial f}{\partial z}\hat{k}$$

$\boxed{\nabla f = f_x\hat{i} + f_y\hat{j} + f_z\hat{k}}$ is called the gradient of f

Find ∇z or $\nabla \omega$

1) $z = 4x - 10y$

$\nabla z = 4\hat{i} - 10\hat{j}$

2) $\omega = \ln\sqrt{x^2+y^2+z^2}$

$$\nabla \omega = \frac{2x}{2\sqrt{x^2+y^2+z^2}}\hat{i} + \frac{2y}{2\sqrt{x^2+y^2+z^2}}\hat{j} + \frac{2z}{2\sqrt{x^2+y^2+z^2}}\hat{k}$$

3) Find the gradient of f at the indicated point

$f(x,y,z) = y\ln(x+y+z)$ at $(-4, 5, 0)$

$$\nabla f = \frac{y}{x+y+z}\hat{i} + \ln(x+y+z)\hat{j} + \frac{y}{x+y+z}\hat{k}$$

$$\nabla f\Big|_{(-4,5,0)} = \underline{\underline{5\hat{i}+5\hat{j}+5\hat{k}}}$$

4) Find the gradient of f at $(-2,-1)$ where $f(x,y) = (x^2+xy)^3$

$$\nabla f = 3(x^2+xy)^2(2x+y)\,\hat{i} + 3(x^2+xy)^2\cdot(x)\,\hat{j}$$

$$\nabla f\Big|_{(-2,-1)} = 3(6)^2(-5)\hat{i} + 3(6)^2(-2)\hat{j}$$

$$= \underline{\underline{-540\hat{i} + -(216)\hat{j}}}$$

Directional Derivative

If $f(x,y)$ is a fn of x & y and if $\bar{u} = u_1\hat{i}+u_2\hat{j}$ is a unit vector then the directional derivative of f at (x_0,y_0) in the direction of $\bar{u}$ is denoted by $D_u f(x_0,y_0)$

$$\boxed{D_u f(x_0,y_0) = \nabla f(x_0,y_0)\cdot u}$$

1) Find the directional derivative at P in the direction of given vector where $f(x,y) = (1+xy)^{3/2}$ $P(8,1)$ $\bar{u} = \frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j}$

$$D_u f(x_0, y_0) = \nabla f(x_0, y_0) \cdot \bar{u}$$

$$= \frac{3}{2}(1+xy)^{\frac{1}{2}} y \Big/_{(8,1)} \hat{i} + \frac{3}{2}(1+xy)^{1/2} x \Big/_{(8,1)} \hat{j}$$

$$\cdot \left(\frac{1}{\sqrt{2}} i + \frac{1}{\sqrt{2}} j\right)$$

$$= \left(\frac{9}{2}\hat{i} + 36\hat{i}\right) \cdot \left(\frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j}\right)$$

$$= \frac{9}{2\sqrt{2}} + \frac{36}{\sqrt{2}} = \frac{81}{2\sqrt{2}}$$

2) Find the directional derivative of f at P in the direction of the vector $\bar{a}$

Q $f(x,y) = y^2 \ln x$ $P(1,5)$ and $\bar{a} = -3\hat{i} + 3\hat{j}$

Ans:- $\|\bar{a}\| = \sqrt{3^2 + 3^2} = 3\sqrt{2}$

$\bar{u} = \frac{-1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j}$

$f_x \Big/_{(1,5)} = \frac{y^2}{x} = 25$ $\qquad f_y \Big/_{(1,5)} = 2y \ln x = 0$

$$D_a f(1,5) = (f_x i + f_y j) \cdot \bar{u}$$

$$= (25i) \cdot \left(-\frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j}\right) = \frac{-25}{\sqrt{2}}$$

3. Find the directional derivative of $f(x,y) = \tan^{-1}\frac{y}{x}$ at the point $(-1,1)$ in the direction $\bar{a} = -\hat{i} - \hat{j}$

Ans:- $f_x = \frac{1}{1+\left(\frac{y}{x}\right)^2} \cdot y \cdot \left(\frac{-1}{x^2}\right)$ $\quad f_y = \frac{1}{1+\left(\frac{y}{x}\right)^2} \cdot \frac{1}{x}$

$f_x(-1,1) = \frac{-1}{2}$ $\quad f_y(-1,1) = \frac{-1}{2}$

$\|\bar{a}\| = \sqrt{1+1} = \sqrt{2}$, $\quad \bar{u} = \frac{-1}{\sqrt{2}}\hat{i} - \frac{1}{\sqrt{2}}\hat{j}$

$\nabla f(-1,1) = -\frac{1}{2}\hat{i} - \frac{1}{2}\hat{j}$

$D_{\bar{u}} f(-1,1) = \left(-\frac{1}{2}\hat{i} - \frac{1}{2}\hat{j}\right) \cdot \left(\frac{-1}{\sqrt{2}}\hat{i} - \frac{1}{\sqrt{2}}\hat{j}\right)$

$= \frac{1}{2\sqrt{2}} + \frac{1}{2\sqrt{2}} = \frac{2}{2\sqrt{2}} = \underline{\underline{\frac{1}{\sqrt{2}}}}$

4. Find the directional derivative of $f(x,y) = \sqrt{xy}\, e^y$ at $P(1,1)$ in the direction of -ve y axis

Ans:- The unit vector corresponding to -y axis is $-\hat{j}$

$\bar{u} = -\hat{j}$

$\nabla f = f_x\hat{i} + f_y\hat{j}$

$= \left(e^y \cdot \frac{1}{2\sqrt{xy}}\, y\right)\hat{i} + \left(\sqrt{xy}\, e^y + \frac{1}{2\sqrt{xy}}\, x e^y\right)\hat{j}$

$$\nabla f(1,1) = \frac{e}{2}\hat{i} + \frac{3e}{2}\hat{j}$$

$$D_u f(1,1) = \nabla f(1,1) \cdot \bar{u}$$

$$= \left(\frac{e}{2}\hat{i} + \frac{3e}{2}\hat{j}\right)(-\hat{j}) = \underline{\frac{-3}{2}e}$$

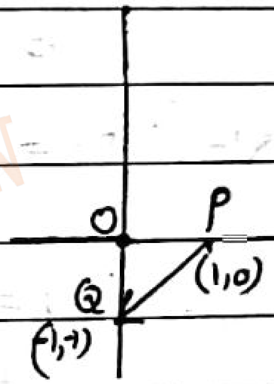
5. Find the directional derivative of $f(x,y) = \frac{x}{x+y}$ at $P(1,0)$ in the direction of $Q(-1,-1)$

$$\overrightarrow{PQ} = \overline{OQ} - \overline{OP}$$

$$= (-\hat{i}-\hat{j}) - (\hat{i}+0\hat{j})$$

$$= -2\hat{i} - \hat{j}$$

which is not a unit vector



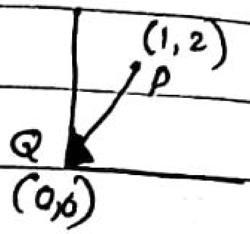
$$\therefore \bar{u} = \frac{\overline{PQ}}{\|PQ\|} = \frac{-2\hat{i}-\hat{j}}{\sqrt{5}} = \frac{-2}{\sqrt{5}}\hat{i} - \frac{1}{\sqrt{5}}\hat{j}$$

$$\nabla f = f_x\hat{i} + f_y\hat{j}$$

$$\nabla f(1,0) = \frac{(x+y)-x}{(x+y)^2}\hat{i} + \frac{-x}{(x+y)^2}\hat{j} = 0\hat{i} - \hat{j} = -\hat{j}$$

$$\nabla f(1,0) \cdot \bar{u} = (-\hat{j}) \cdot \left(\frac{-2}{\sqrt{5}}\hat{i} - \frac{1}{\sqrt{5}}\hat{j}\right) = \underline{\frac{1}{\sqrt{5}}}$$

6. For the above problem, find the directional derivative of f at P(1,2) in the direction of origin



$\overline{PQ} = (0,0) - (1,2)$

$= (-1,-2)$

$= -\hat{i} - 2\hat{j}$ which is not a unit vector

$\|\overline{PQ}\| = \sqrt{5}$

$\bar{u} = \frac{-1}{\sqrt{5}}\hat{i} - \frac{2}{\sqrt{5}}\hat{j}$

$\nabla f(1,2) = \frac{2}{9}\hat{i} - \frac{1}{9}\hat{j}$

$D_u f(1,2) = \nabla f(1,2) \cdot \bar{u}$

$= \left(\frac{2}{9}\hat{i} - \frac{1}{9}\hat{j}\right) \cdot \left(\frac{-1}{\sqrt{5}}\hat{i} - \frac{2}{\sqrt{5}}\hat{j}\right)$

$= \frac{-2}{9\sqrt{5}} + \frac{2}{9\sqrt{5}} = \underline{\underline{0}}$

How to obtain a unit vector from a given vector.

* What is a unit vector?

A vector with magnitude 1 is called a unit vector. (ie norm of the vector = 1 unit)

eg:- ① $\hat{i} \longrightarrow$ unit vector in the direction of x axis

② $\hat{j} \longrightarrow$ " " " y axis

③ $\hat{k} \longrightarrow$ unit vector in the direction of z axis.

④ $\bar{a} = \frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j}$ $\quad \|\bar{a}\| = \sqrt{\left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2} = \sqrt{\frac{1}{2}+\frac{1}{2}} = 1$

$\therefore \bar{a}$ is a unit vector

⑤ Let $\bar{a} = \hat{i} + \hat{j}$

$\|\bar{a}\| = \sqrt{1^2+1^2} = \sqrt{2} \neq 1$

$\therefore \bar{a}$ is not a unit vector.

To make $\bar{a}$, a unit vector divide $\bar{a}$ by its $\|\bar{a}\|$

ie Unit vector $\bar{u} = \frac{\bar{a}}{\|\bar{a}\|} = \frac{\hat{i}+\hat{j}}{\sqrt{2}} = \frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j}$

H.W ⑥ Check whether $3\hat{i} + 2\hat{j}$ is a unit vector or not. If not, form the unit vector.

Ans

$u = \frac{3}{\sqrt{13}}\hat{i} + \frac{2}{\sqrt{13}}\hat{j}$

Check $\leftarrow$

Vector fields:

A vector field in a plane is a function that associates with each point P in the plane a unique vector F(P) parallel to the plane. IIIrly a vector field in 3-space is a function that associates with each point P in 3 space a unique vector F(P) in the space.

A vector field in 2-space is expressed as

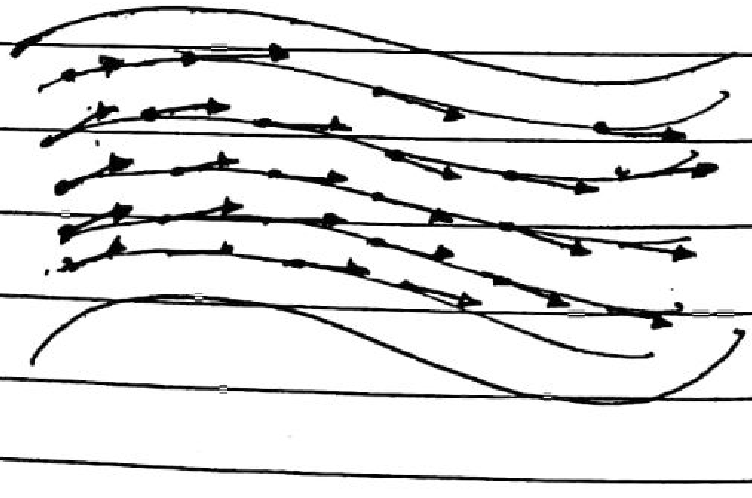
$$F(x,y) = f(x,y)\,\hat{i} + g(x,y)\,\hat{j}$$

In 3 space

$$F(x,y,z) = f(x,y,z)\,\hat{i} + g(x,y,z)\,\hat{j} + h(x,y,z)\,\hat{k}$$

eg:- fluid flow

Imagine a stream in which the water flows horizontally at ~~each point~~ every level, and consider the layer of water at a specific depth. At each point of the layer, the water has a certain velocity, which we can represent by a vector at that point. This association of velocity vectors with points in two dimensional layer is called the velocity field at that layer.



Operations on Vector fields in 3 space

The divergence & Curl of the field

The way in which fluid flows towards or away from a point & The rotational properties of the fluid at a point

Divergence of F & Curl of F

If $F(x,y,z) = f(x,y,z)\,\hat{i} + g(x,y,z)\,\hat{j} + h(x,y,z)\,\hat{k}$ then, we define divergence of F, & curl of F as follows

$$\boxed{\operatorname{div} F = \frac{\partial f}{\partial x} + \frac{\partial g}{\partial y} + \frac{\partial h}{\partial z} = \bar{\nabla}\cdot\bar{F}}$$

$$\boxed{\operatorname{Curl} F = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ f & g & h \end{vmatrix} = \bar{\nabla}\times\bar{F}}$$

The laplacian operator

$$\nabla^2 = \nabla\cdot\nabla = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

Laplace Eqn

$$\nabla^2\phi = 0$$

$$\frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} + \frac{\partial^2\phi}{\partial z^2} = 0$$

1) Find the divergence & curl of $F(x,y,z) = x^2y\,\hat{i} + 2y^3z\,\hat{j} + 3z\,\hat{k}$

Ans:- $\text{div } F = \frac{\partial}{\partial x}x^2y + \frac{\partial}{\partial y}2y^3z + \frac{\partial}{\partial z}3z$

$= 2xy + 6y^2z + 3$

$$\text{Curl } F = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2y & 2y^3z & 3z \end{vmatrix}$$

$= -2y^3\hat{i} - x^2\hat{k}$

2) $F(x,y,z) = x^2\hat{i} - 3\hat{j} + y^2z\hat{k}$ find divergence & curl of F

$\text{div } F = 2x + 2yz$

$\text{Curl } F = z^2\hat{i}$

Find divergence & Curl of the following

3) $F(x,y,z) = 7y^3z^2\hat{i} - 8x^2z^5\hat{j} - 4xy^4\hat{k}$

Ans $\text{div } F = 0$

Curl F

4) $F(x,y,z) = e^{xy}\hat{i} - 2\cos y\,\hat{j} + \sin^2 z\,\hat{k}$

Ans $\text{div } F = ye^{xy} + 2\sin y + 2\sin z\cos z$

$\text{Curl } F = -xe^{xy}\hat{k}$

5) $F(x,y,z) = yz\hat{i} + xy^2\hat{j} + yz^2\hat{k}$

6) $F(x,y,z) = \ln y\,\hat{i} + e^{xyz}\hat{j} + \tan^{-1} z/x\,\hat{k}$

7) S.T the function $z = x^2 - y^2 + 2xy$ satisfies laplace equation $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0$

Ans:- $\frac{\partial z}{\partial x} = 2x + 2y$ $\quad \frac{\partial^2 z}{\partial x^2} = 2$

$\frac{\partial z}{\partial y} = -2y + 2x$ $\quad \frac{\partial^2 z}{\partial y^2} = -2$

$\nabla^2 z = 0$

8. Find $\nabla\cdot(F\times G)$ where $F(x,y,z) = 2x\hat{i} + \hat{j} + 5y\hat{k}$ and $G(x,y,z) = x\hat{i} + y\hat{j} - z\hat{k}$

Ans

$$F\times G = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2x & 1 & 5y \\ x & y & -z \end{vmatrix}$$

$$= \hat{i}(-z-5y^2) - \hat{j}(-2xz-5xy) + \hat{k}(2xy-x)$$

$$\nabla\cdot(F\times G) = \frac{\partial}{\partial x}(-z-5y^2) + \frac{\partial}{\partial y}(2xz-5xy) + \frac{\partial}{\partial z}(2xy-x)$$

$$= +5x$$

9. $\nabla\times(\nabla\times F)$ where $F(x,y,z) = xyz\hat{i} + xy\hat{j}$

Ans:-

$$\nabla\times F = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xyz & xy & 0 \end{vmatrix} = xy\hat{j} + (y-xz)\hat{k}$$

$$\nabla\times(\nabla\times F) = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & xy & y-xz \end{vmatrix} = \hat{i} + z\hat{j} + y\hat{k}$$

#HW
10) Find $\nabla\cdot(\nabla\times F)$, $F(x,y,z) = \sin x\,\hat{i} + \sin(x-y)\,\hat{j} + 2\hat{k}$

Ans $=0$

HW 11) Find $\nabla\cdot(F\times G)$

$F = yz\hat{i} + xz\hat{j} + xy\hat{k}$

$G = xy\hat{i} + xyz\hat{j}$

Ans:- $\nabla\cdot(F\times G) = x^2y$

UQ

12) $\nabla\cdot(\nabla\times F)$ & $\nabla\times(\nabla\times F)$

$F(x,y,z) = e^{xz}\hat{i} + 4xe^{y}\hat{j} - e^{yz}\hat{k}$

$\nabla\cdot(\nabla\times F) = 0$

13) Verify that the radius vector $\bar{r} = x\hat{i} + y\hat{j} + z\hat{k}$ has the stated property provided $r = \|\bar{r}\|^2$

Ans:- 1) Curl $\bar{r} = 0$ 3) div $\bar{r} = 3$

2) $\nabla\|r\| = \dfrac{\bar{r}}{\|r\|}$ 4) $\nabla\dfrac{1}{\|r\|} = \dfrac{-\bar{r}}{\|r\|^3}$

UQ 5) $\nabla f(r) = \dfrac{f'(r)}{\|r\|}\bar{r}$ H.W 6) $\nabla r^n = nr^{n-2}\,\bar{r}$

UQ 7) $\nabla^2 r^n = n(n+1)r^{n-2}$ OR 8) $\nabla\cdot(\nabla r^n) = n(n+1)r^{n-2}$

OR 9) $\text{div}(\text{grad } r^n) = n(n+1)r^{n-2}$

UQ 10) $\nabla\log r = \dfrac{\bar{r}}{r^2}$

Ans ① Curl $\bar{r} = \nabla\times r$

$$= \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x & y & z \end{vmatrix} = i(0) - j(0) + k(0)$$

$$= \underline{\underline{0}}$$

③ $\operatorname{div} \bar{r} = \bar{\nabla} \cdot \bar{r}$

$$= \frac{\partial x}{\partial x} + \frac{\partial y}{\partial y} + \frac{\partial z}{\partial z} = 1+1+1 = 3.$$

② $\nabla \|r\| = \left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}\right)\sqrt{x^2+y^2+z^2}, \quad \|r\| = \sqrt{x^2+y^2+z^2}$

$$= \frac{2x}{2\sqrt{x^2+y^2+z^2}}\hat{i} + \frac{2y}{2\sqrt{x^2+y^2+z^2}}\hat{j} + \frac{2z}{2\sqrt{x^2+y^2+z^2}}\hat{k}$$

$$= \frac{x\hat{i}+y\hat{j}+z\hat{k}}{\sqrt{x^2+y^2+z^2}} = LHS$$

$$\frac{\bar{r}}{\|\bar{r}\|} = \frac{x\hat{i}+y\hat{j}+z\hat{k}}{\sqrt{x^2+y^2+z^2}} = RHS$$

$$\therefore \nabla\|r\| = \frac{\bar{r}}{\|r\|}$$

4) $\nabla \frac{1}{\|r\|} = \left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}\right)\frac{1}{\sqrt{x^2+y^2+z^2}}$

$$= -\frac{1}{2}(x^2+y^2+z^2)^{-3/2} \cdot 2x\,\hat{i} +$$

$$-\frac{1}{2}(x^2+y^2+z^2)^{-3/2} \cdot 2y\,\hat{j} +$$

$$-\frac{1}{2}(x^2+y^2+z^2)^{-\frac{3}{2}} \cdot 2z\,\hat{k}$$

$$= \frac{-(x\hat{i}+y\hat{j}+z\hat{k})}{(x^2+y^2+z^2)^{3/2}} = -\frac{\bar{r}}{\|r\|^3}$$

$r^2 = x^2+y^2+z^2$

$2r \cdot \frac{\partial r}{\partial x} = 2x$

$\frac{\partial r}{\partial x} = \frac{x}{r}, \frac{\partial r}{\partial y} = \frac{y}{r}$

$\frac{\partial r}{\partial z} = \frac{z}{r}$

5) $\nabla f(\bar{r}) = f'(r) \frac{\bar{r}}{\|\bar{r}\|}$

$$\nabla f(\bar{r}) = \frac{\partial}{\partial x} f(r)\hat{i} + \frac{\partial}{\partial y} f(r)\hat{j} + \frac{\partial}{\partial z} f(r)\hat{k}$$

$$= f'(r)\frac{\partial r}{\partial x}\hat{i} + f'(r)\cdot\frac{\partial r}{\partial y}\hat{j} + f'(r)\cdot\frac{\partial r}{\partial z}\hat{k}$$

$$= f'(r)\frac{x}{r}\hat{i} + f'(r)\frac{y}{r}\hat{j} + f'(r)\cdot\frac{z}{r}\hat{k}$$

$$= \frac{f'(r)}{r}\bar{r} = \frac{f'(r)}{\|\bar{r}\|}\bar{r}$$

6) $\nabla r^n = \frac{\partial}{\partial x} r^n\hat{i} + \frac{\partial}{\partial y} r^n\hat{j} + \frac{\partial}{\partial z} r^n\hat{k}$

$$= n r^{n-1}\cdot\frac{x}{r}\hat{i} + n r^{n-1}\frac{y}{r}\hat{j} + n r^{n-1}\cdot\frac{z}{r}\hat{k}$$

$$= n r^{n-2}\bar{r}$$

7) or 8) or 9) $\nabla^2 r^n = \nabla^2(r^n)$

$$= \frac{\partial^2}{\partial x^2} r^n + \frac{\partial^2}{\partial y^2} r^n + \frac{\partial^2}{\partial z^2} r^n$$

$$= \frac{\partial}{\partial x} n r^{n-1}\cdot\frac{x}{r} + \frac{\partial}{\partial y} n r^{n-1}\frac{y}{r} + \frac{\partial}{\partial z} n r^{n-1}\cdot\frac{z}{r}$$

$$= \frac{\partial}{\partial x} n r^{n-2} x + \frac{\partial}{\partial y} n r^{n-2} y + \frac{\partial}{\partial z} n r^{n-2} z$$

$$= n\left[r^{n-2}+(n-2)r^{n-3}\cdot\frac{x}{r}\right)x\Big] +$$

$$n\left[r^{n-2}+(n-2)r^{n-3}\cdot\frac{y}{r}\cdot y\right] +$$

$$n\left(r^{n-2}+(n-2)r^{n-3}\cdot\frac{z}{r}\cdot z\right]$$

$$= n\left[3r^{n-2}+(n-2)r^{n-4}(x^2+y^2+z^2)\right]$$

$$= n\left[3r^{n-2}+(n-2)r^{n-4}\cdot r^2\right]$$

$$= n\left[3r^{n-2}+(n-2)r^{n-2}\right]$$

$$= n r^{n-2}[n+1]$$

$$= \underline{n(n+1)r^{n-2}}$$

Note:- Vector field F is solenoidal (divergence free) if $\nabla\cdot F=0$

ie $\operatorname{div} F=0$

Vector field F is irrotational if $\nabla\times F=0$

OR $\operatorname{Curl} F=0$

The vector field seems like it doesn't rotate

4Q). Determine 'a' so that

$(x+3y)\hat{i} + (y-2x)\hat{j} + (x+az)\hat{k}$ is solenoidal.

Ans: If F is solenoidal $\nabla \cdot F = 0$

$\nabla \cdot F = 1+1+a = 0$

$= 2+a = 0$

$\therefore \underline{\underline{a = -2}}.$

4Q) Find the values of a, b, c s.t

$F = (axy + bz^3)\hat{i} + (3x^2 - zc)\hat{j} + (3xz^2 - y)k$

may be irrotational.

Ans: F is irrotation, then Curl $F = 0$

$$\nabla \times F = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ axy+bz^3 & 3x^2-zc & 3xz^2-y \end{vmatrix}$$

$= \hat{i}(-1+c) - \hat{j}(3z^2 - 3bz^2) + \hat{k}(6x - ax)$

$= (c-1)\hat{i} + 3z^2(b-1)\hat{j} + (6x-ax)\hat{k} = 0$

$\therefore c-1 = 0 \Rightarrow \underline{\underline{c=1}}$ $\quad 6x - ax = 0$

$b-1 = 0 \Rightarrow \underline{\underline{b=1}}$ $\quad \therefore \underline{\underline{a=6}}$

Line Integral.

Any integral which is to be evaluated along a curve is called line integral. If C is a curve in the 3 space that is smoothly parametrised by

$\bar{r}(t) = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}$ $a \leq t \leq b$

then

$$\int_a^b f(x,y,z)\, ds = \int_a^b f(x(t), y(t), z(t)) \cdot \|\bar{r}'(t)\| d$$

1. Evaluate the integral $\int_C 1+xy^2\, ds$
   a) $C: r(t) = t\hat{i} + 2t\hat{j}$ $0 \leq t \leq 1$
   b) $C: r(t) = (1-t)\hat{i} + (2-2t)\hat{j}$ $0 \leq t \leq 1$

Ans

$$\int_C 1+xy^2\, ds = \int_0^1 f(x(t), y(t))\, \|r'(t)\|\, dt$$

$$\boxed{\begin{aligned} r'(t) &= \hat{i} + 2\hat{j} \\ \|\bar{r}'(t)\| &= \sqrt{1+4} = \sqrt{5} \end{aligned}}$$

$x = t$

$y = 2t$

$$= \int_0^1 \left(1 + t\,(2t)^2\right)\sqrt{5}\, dt$$

$$= \sqrt{5}\left(t + \frac{4t^4}{4}\right)_0^1 = \sqrt{5}(1+1) = \underline{\underline{2\sqrt{5}}}$$

b) $$\int_C (1+xy)\,ds = \int_0^1 1 + (1-t)(2-2t)^2 \sqrt{5}\,dt$$

$$= \int_0^1 \left(1 + 4(1-t)^3\right)\sqrt{5}\,dt$$

$$= \sqrt{5}\left(t + \frac{4(1-t)^4}{-4}\right)_0^1$$

$$= \sqrt{5}\left(1 + 0 - 0 - {}^-1\right) = \underline{\underline{2\sqrt{5}}}$$

2) Evaluate $\int_C 2xy\,dx + (x^2+y^2)\,dy$ along the circular arc C given by
$x = \cos t$, $y = \sin t$, $0 \le t \le \pi/2$

Ans:- $$\int_0^{\pi/2} (2\cos t \cdot \sin t) -\sin t\,dt + \cos t\,dt$$

$$= \int_0^{\pi/2} -2\sin^2 t\cos t + \cos t \cdot dt$$

$$= \left(\frac{-2\sin^3 t}{3}\right)_0^{\pi/2} + (\sin t)_0^{\pi/2}$$

$$= -\frac{2}{3}\left(\sin^3\frac{\pi}{2} - \sin^3 0\right) + \left(\sin\frac{\pi}{2} - \sin 0\right)$$

$$= -\frac{2}{3} + 1 = \underline{\underline{+1/3}}$$

11th 3) Evaluate $\int_C (3x^2 + y^2)\,dx + 2xy\,dy$ along the circular arc given by $x = \cos t,\ y = \sin t \ (0 \le t \le \pi/2)$.

Ans $= -1$ Ans in last page

Anton.

4) Evaluate the line integral $\int_C (xy + z^3)\,ds$ from $(1,0,0)$ to $(-1,0,\pi)$ along the helix C that is represented by the parametric equations $x = \cos t,\ y = \sin t$ $z = t \ (0 \le t \le \pi)$

Ans:- $\bar{r}(t) = \cos t\,\hat{i} + \sin t\,\hat{j} + t\,\hat{k}$

$\bar{r}'(t) = -\sin t\,\hat{i} + \cos t\,\hat{j} + \hat{k}$

$\|\bar{r}'(t)\| = \sqrt{\cos^2 t + \sin^2 t + 1} = \sqrt{2}$

$$\int (xy + z^3)\,ds = \int_0^{\pi} (\cos t \sin t + t^3)\sqrt{2}\,dt$$

$$= \underline{\underline{\frac{\sqrt{2}\,\pi^4}{4}}}$$

5) In each part evaluate the integral $\int_C (5x+2y)\,dx + (2x-y)\,dy$ along the stated

(a) The line segment from (0,0) to (1,1)

(b) The parabolic arc $y=x^2$ from (0,0) to (1,1)

HW (c) The curve $y=\sin \frac{\pi x}{2}$ from (0,0) to (1,1)

(d) The curve $y=x^3$ from (0,0) to (1,1)

Ans (a) The eqn of the line segment from (0,0) to (1,1) is

$$\frac{x-0}{0-1} = \frac{y-0}{0-1} \Rightarrow x=y \qquad dx=dy$$

$$I = \int_C (5x+2y)\,dx + (2x-y)\,dy$$

$$\int_0^1 (5x+2x)\,dx + (2x-x)\,dx$$

$$= \int_0^1 8x\,dx$$

$$= 8\frac{x^2}{2}\Big]_0^1 = \underline{\underline{4}}$$

(b) $y=x^2$, $dy = 2x\,dx$

$$\int_0^1 \left(5x+2x^2\right)dx + \left(2x - x^2\right)2x\,dx$$

curves $= \int_0^1 5x + 2x^2 + 4x^2 - 2x^3 \, dx$

$$= \int_0^1 -2x^3 + 6x^2 + 5x \, dx$$

$$= \left.\frac{-2x^4}{4} + \frac{6x^3}{3} + \frac{5x^2}{2}\right)_0^1$$

$$= -\frac{1}{2} + 2 + \frac{5}{2} = 2 + 2 = \underline{4}$$

6. Evaluate the line integral w.r.s s along the curve C.

$$\int_C \frac{1}{1+x}\,ds, \quad C: \bar{r}(t) = t\hat{i} + \frac{2}{3}t^{3/2}\hat{j} \quad 0 \leq t \leq 4.$$

$$\bar{r}'(t) = \hat{i} + \frac{2}{3} \times \frac{2}{3}.t^{1/2}\hat{j}$$

$$\|\bar{r}'(t)\| \sqrt{1+t}$$

$$\int_0^4 \frac{1}{1+t}\sqrt{1+t}\,dt$$

$$= \int_0^4 \frac{1}{\sqrt{1+t}}\,dt = \left(2\sqrt{1+t}\right)_0^4$$

$$= \underline{2(\sqrt{5}-1)}$$

7. $\int_C 3xyz\, ds$ $\quad C: x=t,\ y=t^2,\ z=\frac{2}{3}t^3,\ 0\le t\le$

Ans $= \frac{46}{63}$

8. $\int_C (x+3y)\,dx+(x-y)\,dy$ $\quad C: x=2\cos t;\ y=4\sin t,\ 0\le t\le \pi/4$

Ans $= 3-2\pi$

9. $\int_C -y\,dx+x\,dy$ $\quad C: y^2=3x$ from $(3,3)$ to $(0,0)$

Ans $= 3$

3. $\int_0^{\pi/2} (3\cos^2 t+\sin^2 t)\,-\sin t\,dt + 2\cos t\sin t\,\cos t\,dt$

$= \int_0^{\pi/2} -3\cos^2 t\sin t - \sin^3 t + 2\cos^2 t\sin t\ dt$

$= \int_0^{\pi/2} -\cos^2 t\sin t - \sin^3 t\ dt$

$= \int_0^{\pi/2} -\sin t\,(\cancel{\cos^2 t+\sin^2 t})^{1}\,dt$

$= (\cos t)_0^{\pi/2} = \cos\frac{\pi}{2} - \cos 0 = \underline{\underline{-1}}$

Work as a line integral.

Suppose that under the influence of a continuous force field F a particle moves along a smooth curve C and the C is oriented in the direction of motion of the particle. Then the work performed by the force field on the particle is

$$\text{Work done} = \int_C F \cdot dr$$

Note: 1.

If $\bar{r} = x\hat{i} + y\hat{j} + z\hat{k}$ (position vector)

then $d\bar{r} = dx\hat{i} + dy\hat{j} + dz\hat{k}$

$$\therefore \int_C F \cdot dr = \int_C (f\hat{i} + g\hat{j} + h\hat{k}) \cdot (dx\hat{i} + dy\hat{j} + dz\hat{k})$$

$$\boxed{\int_C F \cdot dr = \int_C f\,dx + g\,dy + h\,dz}$$

Note: 2

If $\bar{r}(t) = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}$, $dr = \bar{r}'(t)\,dt$

then $\int_C F \cdot dr = \int_C F \cdot \bar{r}'(t)\,dt$

$$\boxed{\int_C F \cdot dr = \int_C F(x(t), y(t), z(t)) \cdot \bar{r}'(t)\,dt}$$

1Q) Evaluate $\int F\cdot dr$ where $F(x,y) = \sin x\,\hat{i} + \cos x\,\hat{j}$ where C is the curve $\bar{r}(t) = \pi\hat{i} + t\hat{j}$ $0 \le t \le 2$

Ans $\int_C F\cdot dr = \int_0^2 (\sin x\,\hat{i} + \cos x\,\hat{j})\cdot\hat{j}\,dt$ $\quad x=\pi,\ y=t$

$$\int_C F\cdot r'(t)dt = \int_0^2 (\sin\pi\,\hat{i} + \cos\pi\,\hat{j})\cdot\hat{j}\,dt$$

$$= \int_0^2 \cos\pi\,dt = \int_0^2 \cancel{\sin t} -1\,dt$$

$$= (t)_0^2 = \underline{-2}$$

1Q) 2) Find the work done by the force field $F(x,y,z) = (x^2+xy)\hat{i} + (y - x^2y)\hat{j}$ on the particle that moves along the curves C: $x = t,\ y = \frac{1}{t},\ 1 \le t \le 3$

Ans

$$\int_C F\cdot dr = \int_C F(t)\cdot r'(t)\,dt$$

$$r(t) = t\hat{i} + \frac{1}{t}\hat{j}$$

$$\bar{r}'(t) = \hat{i} + -\frac{1}{t^2}\hat{j}$$

$$\int_C F\cdot dr = \int_1^3 \left((t^2+1)\hat{i} + \left(\frac{1}{t} - t\right)\hat{j}\right)\cdot\left(\hat{i} + \frac{-1}{t^2}\hat{j}\right)dt$$

$$\int_1^3 (t^2+1) - \frac{1}{t^2}\left(\frac{1}{t} - t\right) dt$$

$$= \int_1^3 t^2+1 - \frac{1}{t^3} + \frac{1}{t}\, dt$$

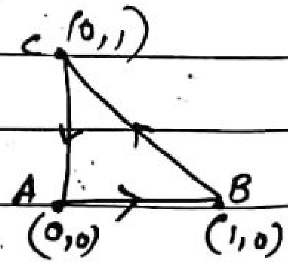
$$= \left(\frac{t^3}{3} + t - \frac{t^{-2}}{-2} + \ln t\right)_1^3$$

$$= 9+3+\frac{1}{18} + \ln 3 - \frac{1}{3} - 1 \quad - \ln 1$$

$$= \frac{92}{9} + \ln 3$$

$$=$$

3. Evaluate $\int_C F \cdot dr$ where $F(x,y) = y\hat{i} - x\hat{j}$ along the Δle joining the vertices $(0,0)$, $(1,0)$ & $(0,1)$



Ans $\int_C F \cdot dr = \int_{AB} F \cdot dr + \int_{BC} F \cdot dr + \int_{AC} F \cdot dr$

Along AB

$y = 0 \quad dy = 0$

$dr = dx\hat{i} + dy\hat{j} = dx\hat{i}$

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$$\int_{AB} F\cdot d\gamma = \int (y\hat{i} - x\hat{j})\cdot(dx\hat{i} + dy\hat{j})$$

$$= \int_0^1 (-x\hat{j})\cdot dx\hat{i} = \underline{0}$$

$$\int_{BC} F\cdot d\gamma = \int (y\hat{i} - x\hat{j})\cdot(dx\hat{i} + dy\hat{j})$$

Eqn of BC

$$\frac{x-0}{0-1} = \frac{y-1}{1-0}$$

$$-x = y - 1$$

$$y + x = 1$$

$$x + y = 1$$

$$x = 1 - y$$

$$dx = -dy$$

$$= \int (y\hat{i} - (1-y)\hat{j})\cdot(-dy\hat{i} + dy\hat{j})$$

$$= \int_0^1 -y\,dy - (1-y)\,dy$$

$$= \int_0^1 -1\,dy = -y\Big]_0^1 = \underline{-1}$$

$$\int_{BC} F\cdot d\gamma = \int (y\hat{i} - x\hat{j})\cdot(dx\hat{i} + dy\hat{j})$$

$$x = 0$$

$$dx = 0$$

$$= \int y\,dx - x\,dy$$

$$= \int 0 = 0$$

$$\int F\cdot d\gamma = \underline{\underline{-1}}$$

UQ

4. Find the work done by a force field $F(x,y,z) = xy\hat{i} + yz\hat{j} + xz\hat{k}$ on a particle that moves along a curve
$C: r(t) = t\hat{i} + t^2\hat{j} + t^3\hat{k} \quad 0 \le t \le 1$

Ans $\int_C F \cdot dr = \int_C F(t) \cdot r'(t)\, dt$

$= \int_0^1 \left(t \cdot t^2 \hat{i} + t^2 \cdot t^3 \hat{j} + t t^3 \hat{k}\right) \cdot \left(\hat{i} + 2t\hat{j} + 3t^2 \hat{k}\right) dt$

$= \int_0^1 t^3 + 2t^6 + 3t^6\, dt$

$= \left(\frac{t^4}{4} + \frac{5t^7}{7}\right)_0^1 = \frac{1}{4} + \frac{5}{7} = \underline{\underline{\frac{27}{28}}}$

UQ

5. Find the work done by the force field $F = (x+y)\hat{i} + xy\hat{j} - z^2\hat{k}$ along the line segment from $(0,0,0)$ to $(1,3,1)$ and then to $(2,-1,5)$

Ans :- Let $O(0,0,0)$, $A(1,3,1)$, $B(2,-1,5)$

Along OA $(0,0,0)$ to $(1,3,1)$

$\frac{x-0}{0-1} = \frac{y-0}{0-3} = \frac{z-0}{0-1} \Rightarrow -x = -\frac{y}{3} = -z$

Along (0,0,0) to (1,3,1) $\quad x = z, \; y = 3z/1$

$$\int_{OA} \left(4z\hat{i} + 3z^2\hat{j} - z^2\hat{k}\right)\cdot\left(\hat{i} + 3\hat{j} + \hat{k}\right) dz$$

$$= \int_0^1 \left(4z + 9z^2 - z^2\right) dz$$

$$= \int_0^1 4z + 8z^2 \, dz = \left(\frac{4z^2}{2} - 8\frac{z^3}{3}\right)_0^1$$

$$= 2 + \frac{8}{3} = \frac{14}{3}$$

Along AB (1,3,1) to (2,-1,5)

$$\frac{x-1}{2-1} = \frac{y-3}{-1-3} = \frac{z-1}{5-1}$$

$$x-1 = \frac{y-3}{-4} = \frac{z-1}{4} = t$$

$x = 1$, $t = 0$

$x = t+1, \; y = 3-4t, \; z = 4t+1$

$dx = dt, \; dy = -4dt, \; dz = 4dt$

$x = 2$, $t = 1$

$$\int_0^1 \left((4-3t)\hat{i} + (t+1)(3-4t)\hat{j} - (4t+1)^2\hat{k}\right)\cdot\left(\hat{i} - 4\hat{j} + 4\hat{k}\right) dt$$

$$= \int_0^1 (4-3t) - 4(-4t^2+3-t) - 4(4t+1)^2 \, dt = \frac{-87}{2}$$

$\frac{14}{3} - \frac{87}{2} =$ Ans $= \frac{-233}{6}$

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6. Find the work done by $F = xy\hat{i} + x^3\hat{j}$ on a particle that moves along the curve $y^2 = x$ from $(0,0)$ to $(1,1)$

$$= \int_C F \cdot dr$$

$$= \int_C xy\,dx + x^3\,dy$$

$$= \int_C y^3 \cdot 2y\,dy + y^6\,dy = \frac{19}{35}$$

7. Evaluate $\int_C F \cdot dr$ $C: r(t) = 2\cos t\,\hat{i} + 2\sin t\,\hat{j}$
$0 \le t \le \pi$

where $F(x,y) = x^2\hat{i} + xy\hat{j}$

$W = 0$

Independence of path; conservative vector field

The parametric curve C in a work integral is called the path of integration. If the force field F is conservative then the work done on a particle doesn't depend on the path C but only at end points.

Consider F is a conservative vector field i.e. $F = \nabla\phi$

Then the following results follows

1) In 2D plane, $\bar{F} = \nabla\phi$ iff $\frac{\partial f}{\partial y} = \frac{\partial g}{\partial x}$

2) If 3D space $F = \nabla\phi$ iff $\nabla \times F = 0$
Curl $F = 0$

3) $W = \int_C F \cdot d\gamma = 0$ for any closed curve C

4) $W = \int_C F \cdot d\gamma$ is independent of path.

(5) If C starts from (x_0, y_0) to (x_1, y_1)

$$\int_C F \cdot dr = \phi(x_1, y_1) - \phi(x_0, y_0)$$

is called fundamental theorem of line integral.

Note:-

If ϕ is the potential function, then $-\phi$ is the potential energy.

Conservative field & Potential function

A vector field F is said to be conservative in a region, if it the gradient field for some function ϕ in that region.

ie $F = \nabla\phi$

Then ϕ is called potential function for F in the region.

Problems

Q Confirm that ϕ is a potential function for F in some region &

1. $\phi(x,y) = \tan^{-1}xy$, $F(x,y) = \frac{y}{1+x^2y^2}\hat{i} + \frac{x}{1+x^2y^2}\hat{j}$

$\nabla\phi = F$ $\therefore$ ϕ is the potential fn for F

HW

Q.

2 $\phi(x, y, z) = x^2 - 3y^2 + 4z^3$, $Z = 2x\hat{i} - 6y\hat{j} + 12z^2\hat{k}$

3\. $\phi(x, y, z) = 2y^2 + 3x^2y - xy^4$, $F = 6xy - y^4\,\hat{i} + (4y + 3x)\hat{j} - 4xy^3\hat{j}$

Determine whether F is conservative, if so find a potential function for it.

1) $F(x,y) = 2x\hat{i} + 2y\hat{j}$ (2D)

$f = 2x$ $\quad$ $g = 2y$

$\frac{\partial f}{\partial y} = 0$ $\quad$ $\frac{\partial g}{\partial x} = 0$

$\frac{\partial f}{\partial y} = \frac{\partial g}{\partial x}$ [By the 1st point (in 2D plane)]

$\therefore$ F is conservative (Independent of path)

Since F is conservative $F = \nabla \phi$

$2x\hat{i} + 2y\hat{j} = \nabla \phi$

$2x\hat{i} + 2y\hat{j} = \frac{\partial \phi}{\partial x}\hat{i} + \frac{\partial \phi}{\partial y}\hat{j}$

$$\boxed{\begin{aligned}\nabla &\equiv \frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k} \\ \nabla\phi &= \frac{\partial \phi}{\partial x}\hat{i} + \frac{\partial \phi}{\partial y}\hat{j} + \frac{\partial \phi}{\partial z}\hat{k}\end{aligned}}$$

Equate the i^{th} components on both sides.

$\frac{\partial \phi}{\partial x} = 2x$

integrate w.r.t x.

$\phi = \frac{2x^2}{2} + C_1$

$\phi = x^2 + C_1$ —— ①

Equate the j^{th} components

$\frac{\partial \phi}{\partial y} = 2y$

integrate w.r.t y.

$\phi = \frac{2y^2}{2} + C_2$

$\phi = y^2 + C_2$ —— ②

$\therefore \phi = x^2 + y^2 + C$ $\xrightarrow{}$ constant together. from ① & 2

is the potential function

2) $F(x,y) = x^2y\,\hat{i} + 6xy^2\,\hat{j}$ (2D) $f = x^2y$

$g = 6xy^2$

$\frac{\partial f}{\partial y} = x^2$, $\frac{\partial g}{\partial x} = 6y^2$

$\frac{\partial f}{\partial y} \neq \frac{\partial g}{\partial x}$ $\therefore$ F is not conservative (Not independent of path)

$\therefore$ There doesn't exist a potential function.

3) $F(x,y) = 2xy^3\,\hat{i} + (1+3x^2y^2)\,\hat{j}$ (2D)

$f = 2xy^3$ $g = 1+3x^2y^2$

$\frac{\partial f}{\partial y} = 6xy^2$ $\frac{\partial g}{\partial x} = 6xy^2$

$\frac{\partial f}{\partial y} = \frac{\partial g}{\partial x}$ $\therefore$ F is conservative and hence work done is independent of path.

Since F is conservative $F = \nabla\phi$

$2xy^3\,\hat{i} + (1+3x^2y^2)\,\hat{j} = \frac{\partial \phi}{\partial x}\,\hat{i} + \frac{\partial \phi}{\partial y}\,\hat{j}$

Equate the ith components

$\frac{\partial \phi}{\partial x} = 2xy^3$

integrate w.r.t x (y keeping const)

$\phi = 2y^3 \cdot \frac{x^2}{2} + C_1$

$\phi = x^2y^3 + C_1$ —— ①

Equate the jth components

$\frac{\partial \phi}{\partial y} = 1+3x^2y^2$

integrate w.r.t y (keeping x constant)

$\phi = y + 3x^2 \cdot y^3/3 + C_2$

$\phi = y + x^2y^3 + C_2$ —— ②

x^2y^3 appears in both eqns

$\therefore$ Take x^2y^3 once while writing final answer

Write only one constant

from ① & ② $\therefore$ $\phi = x^2y^3 + y + C$

4) $F(x,y) = (\cos y + y\cos x)\hat{i} + (\sin x - x\sin y)\hat{j}$ (2D)

$f = \cos y + y\cos x$ $\quad$ $g = \sin x - x\sin y$

~~$\frac{\partial f}{\partial x} = 0 + -y\sin x$~~ $\quad$ ~~$\frac{\partial g}{\partial y} = 0 - x\cos$~~

$\frac{\partial f}{\partial y} = -\sin y + \cos x$ $\quad$ $\frac{\partial g}{\partial x} = \cos x - \sin y$.

$\frac{\partial f}{\partial y} = \frac{\partial g}{\partial x}$ $\quad \therefore$ F is conservative and hence

$F = \nabla\phi$

$(\cos y + y\cos x)\hat{i} + (\sin x - x\sin y)\hat{j} = \frac{\partial\phi}{\partial x}\hat{i} + \frac{\partial\phi}{\partial y}\hat{j}$

Equate the i^{th} & j^{th} Components

$\frac{\partial\phi}{\partial x} = \cos y + y\cos x$

$\int$ w.r.t x (y keep const).

$\phi = \boxed{\cos y \cdot x} + y\cdot\sin x + C_1 \longrightarrow (1)$

$\frac{\partial\phi}{\partial y} = \sin x - x\sin y$

$\int$ w.r.t y (keep x constant)

$\phi = \sin x\cdot y + \boxed{x\cdot\cos y} + C_2 \longrightarrow (2)$

$\therefore \phi = x\cos y + y\sin x + C$ $\quad$ from (1) & (2)

is the potential function.

UQ 5) $F(x,y,z) = (2xy+z^3)\hat{i} + x^2\hat{j} + 3xz^2\hat{k}$ (3D problem)

Check whether F is conservative, if so find the potential function...

Ans:- Since there are 3 components ($\hat{i}, \hat{j}$ & $\hat{k}$) use the 2nd formula.

ie Curl $F = \nabla \times F = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2xy+z^3 & x^2 & 3xz^2 \end{vmatrix}$

$= \hat{i}\left(\frac{\partial}{\partial y}3xz^2 - \frac{\partial}{\partial z}x^2\right) - \hat{j}\left(\frac{\partial}{\partial x}3xz^2 - \frac{\partial}{\partial z}(2xy+z^3)\right) + \hat{k}\left(\frac{\partial x^2}{\partial x} - \frac{\partial}{\partial y}(2xy+z^3)\right)$

$= \hat{i}(0-0) - \hat{j}(3z^2 - 3z^2) + \hat{k}(2x-2x) = 0$

Since Curl $F = 0$, F is a conservative vector field.

$\therefore$ $F = \nabla\phi$. ie, $(2xy+z^3)\hat{i} + x^2\hat{j} + 3xz^2\hat{k} = \frac{\partial\phi}{\partial x}\hat{i} + \frac{\partial\phi}{\partial y}\hat{j} + \frac{\partial\phi}{\partial z}\hat{k}$

$\frac{\partial\phi}{\partial x} = 2xy+z^3$

$\int$ w.r.t x (keep y & z constant)

$\phi = 2y\frac{x^2}{2} + z^3x + C_1$

$\phi = x^2y + \boxed{xz^3} + C_1 \rightarrow (1)$

$\frac{\partial\phi}{\partial y} = x^2$

$\int$ w.r.t y (keep x,z const)

$\phi = x^2y + C_2$

$\phi = x^2y + C_2 \rightarrow (2)$

$\frac{\partial\phi}{\partial z} = 3xz^2$

$\int$ w.r.t z (keep x & y const)

$\phi = 3x\cdot\frac{z^3}{3} + C_3$

$\phi = \boxed{xz^3} + C_3 \rightarrow (3)$

$\therefore$ $\phi = x^2y + xz^3 + C$ from (1), (2) & (3)

6) S.T the integral is independent of path and find its value

$$\int_{(1,2)}^{(4,0)} 4y\,dx + 4x\,dy.$$ (2D problem)

Ans:- The given integral is $W = \int F \cdot dr$.

where $F(x,y) = 4y\,\hat{i} + 4x\,\hat{j}$

$f = 4y \qquad g = 4x$

$\frac{\partial f}{\partial y} = 4 \qquad \frac{\partial g}{\partial x} = 4$

$\frac{\partial f}{\partial y} = \frac{\partial g}{\partial x}$, $\therefore F$ is conservati

$\therefore$ integral is independent of the path. Integral depends only on the end points

$\therefore F = \nabla \phi$

$4y\hat{i} + 4x\hat{j} = \frac{\partial \phi}{\partial x}\hat{i} + \frac{\partial \phi}{\partial y}\hat{j}$

$\frac{\partial \phi}{\partial x} = 4y$	$\frac{\partial \phi}{\partial y} = 4x$
$\int$ wrt x	$\int$ w.r.t y
$\phi = 4yx + C_1$	$\phi = 4xy + C_2$

$\therefore \phi = 4xy + C$ is the potential fn.

$$\int_{(1,2)}^{(4,0)} 4y\,dx + 4x\,dy = \phi(4,0) - \phi(1,2)$$
$$= 4 \times 4 \times 0 + C - [4(1 \times 2) + C]$$
$$= C - 8 - C = \underline{-8}$$ is the work done

H.W

1) S.T the line integral $\int_C y\sin x\,dx - \cos x\,dy$ is independent of path. Evaluate the above line integral along the line segment from $(0,1)$ to $(\pi,-1)$.

Ans:- F is conservative

prove it

$\phi = -y\cos x + C$

$$\int_{(0,1)}^{(\pi,-1)} y\sin x\,dx - \cos x\,dy = \phi(\pi,-1) - \phi(0,1) = 0$$

2) Confirm that the force field F is conservative and then find the workdone by the force field on a particle moving along an arbitrary smooth curve in the region from P to Q, F(x,y).

(a) $F(x,y) = xy^2\,\hat{i} + x^2y\,\hat{j}$ $\quad P(2,2) \quad Q(0,0)$

Ans :- $\phi = \frac{x^2y^2}{2} + C$, $W = -8$

(b) $F(x,y) = ye^{xy}\,\hat{i} + xe^{xy}\,\hat{j}$ $\quad P(-1,1),\ Q(2,1)$

Ans:- $\phi = e^{xy} + C$ $\quad W = e^2 - e^{-1}$

3. H.W S.T $\int_A^B (2xy+z^3)\,dx + x^2 dy + 3xz^2 dz$ is independent of path joining A & B.

Ans:- It is enough to ~~calculate~~ show that Curl $F = 0$

H.W

4) $F = (6xy+z^3)\hat{i} + (3x^2-z)\hat{j} + (3xz^2-y)\hat{k}$ is conservative. Find the scalar potential function

Ans:- S.T Curl $F = 0$

$$\phi = 3x^2y + xz^3 - zy + C$$